Programming with Generative AI

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Lecture 9:

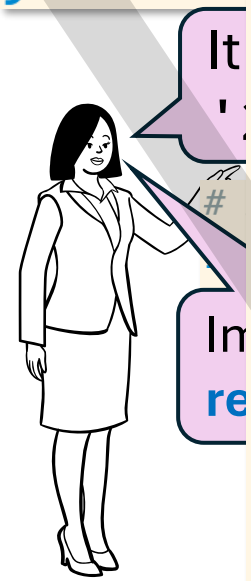
- Lists and Tuples
- Exceptions

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A CSR Initiative by
kotak
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Modify this function

```
# Step # Step 3: Calculate the next date
day = iif day < max_day:
month =     day += 1
year = else:
        day = 1
        if month < 12:
            month += 1
        else:
            month = 1
            year += 1
```



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Ask Clarifying Questions!

```
# Step 1: Extract the day, month, and year
day = int(date[:2])
month = int(date[3:5])
year = int(date[6:])
```

```
# Step 4: Return as str
return f'{day}/{month}/{year}'
```

- What should be output for `'27/02/2025'`?
  - `'28/02/2025'` or `'28/2/2025'`
- What should be output for `'28/02/2025'`?
  - `'01/03/2025'` or `'1/3/2025'`
- What should be output for: CodeCheck
  - `'01/13/2025'` or `'1-3-25'` or `'a/b/c'` or ...

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Splitting a string

```
>>> date = '27/02/2025'
>>> type(date)
    <class 'str'>
>>> date = date.split('/')
>>> id(date)
    1509715255232
>>> id(date[0])
    1509760294064
```

Recall: A **`str`** is **immutable**
In contrast, a **`list`** is **mutable**

Visualize on PythonTutor (switch to non-default visualization)

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Critique this suggestion

```
>>>date = 'a/b/c'
split('/')

ost recent call last):
 invalid literal for
```

```
def next(date: str) -> str:
    """Calculate the next date.
    >>> next('22/10/2008')
    '23/10/2008'
    >>> next('a/b/c')
    Traceback (most recent call last):
    ...
    ValueError: Illegal format: a/b/c
    """
```

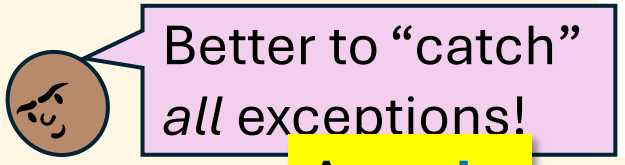
'a/b/c' ✗

Basics of Exceptions

- Our client wants this behaviour:

```
next('01/13/2025') should raise this exception:
ValueError('Illegal format: 01/13/2025')
```

```
# Step 1: Extract the day, month, and year
date = date.split('/')
try:
    day = int(date[0])
    month = int(date[1])
    year = int(date[2])
except:
    raise ValueError(f'Illegal format: {date}')
```



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Tuples: Immutable sequences

```
>>> iisc = (13.0171, 77.5659)
>>> type(iisc)
    <class 'tuple'>
>>> len(iisc)
    2
>>> iisc[0]
    13.0171
>>> iisc[-1]
    77.5659
>>> iisc[1:-1]
```

Can be more efficient than lists

```
    TypeError: 'tuple' object does not support item assignment
```

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Sequence unpacking

- Recall:

```
1  # Step 1: Initialize x and y
2  x = 7.5
3  y = -12
4
5  # Step 2: Swap them without a temporary variable
6  x, y = (y, x)
```

```
# Step 1: Extract the day, month, and year
try:
    day, month, year = date.split('/')
    day = int(day)
    month = int(month)
    year = int(year)
except (ValueError, TypeError):
    raise ValueError(f'Illegal format: {date}')
```

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Summary

- A powerful built-in datatype: **`list`**

  Mutable sequences (can be **heterogeneous**)

  - Negative indexing, Slicing, **`in`**
  - Many methods and functions: **`.sort()`**, **`.append()`**, **`.extend()`**, **`.pop()`**, **`.count()`**, **`max()`**, **`min()`**, …

- Immutable variant: **`tuple`**

- Exceptions in Python:

```
try:
    # potential exception(s)
except (...):
    # handle exception(s)
```

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